

OUTPUTS

UPLOADED FILES

```
['.config', 'ball_014_003.wav', 'ir_007_002.wav', 'ball_007_001.wav', '192.mat', '189.mat',
'ball_014_010.wav', 'or_007_008.wav', 'ball_021_006.wav', 'or_014_002.wav', 'ir_007_007.wav',
'174.mat', '229.mat', 'ball_021_009.wav', 'Second_Part.csv', '203.mat', 'ir_014_008.wav', 'or_007_010.wav',
'ball_014_006.wav', '110.mat', 'normal_008.wav', 'or_007_001.wav', 'First_Part_Small.csv',
'normal_003.wav', 'or_014_006.wav', 'ir_007_008.wav', 'ir_021_004.wav', 'or_014_008.wav',
'or_021_008.wav', '112.mat', '123.mat', 'ball_007_004.wav', '111.mat', 'or_014_003.wav', 'ir_014_003.wav',
'or_021_002.wav', 'normal_001.wav', 'or_014_004.wav', 'ir_014_004.wav', 'ball_007_006.wav', '109.mat',
'ir_014_010.wav', 'ir_014_001.wav', 'or_014_009.wav', 'or_021_004.wav', 'ir_007_004.wav',
'or_014_005.wav', '122.mat', 'ball_021_010.wav', 'ir_007_001.wav', '190.mat', 'ball_021_001.wav',
'or_007_004.wav', 'normal_002.wav', 'ir_007_009.wav', 'ir_014_005.wav', 'or_021_001.wav',
'or_007_002.wav', 'ball_007_009.wav', 'ball_021_007.wav', 'ir_021_006.wav', 'normal_006.wav',
'normal_009.wav', 'ir_014_006.wav', 'ir_014_002.wav', 'ir_007_006.wav', 'ir_007_003.wav',
'ball_007_008.wav', 'ball_007_005.wav', 'normal_004.wav', 'or_007_009.wav', 'ir_021_007.wav',
'ir_021_008.wav', 'or_021_007.wav', 'ball_021_002.wav', 'ball_021_008.wav', 'ball_007_010.wav',
'or_014_007.wav', 'ball_014_002.wav', 'ir_007_010.wav', 'normal_010.wav', 'or_021_009.wav',
'or_014_001.wav', 'ball_007_007.wav', 'Third_Part.csv', 'normal_007.wav', 'ir_007_005.wav',
'ball_007_003.wav', 'ball_014_007.wav', '136.mat', '241-2.mat', 'ball_021_005.wav', 'or_007_007.wav',
'ir_021_005.wav', 'or_007_005.wav', '124.mat', 'ball_014_004.wav', 'or_021_006.wav',
'ball_014_005.wav', 'or_007_006.wav', 'ir_021_010.wav', 'ir_021_009.wav', 'or_014_010.wav', '226.mat',
'ir_014_009.wav', 'normal_005.wav', 'ball_021_004.wav', '125.mat', 'ir_021_002.wav', 'ir_021_001.wav',
'ball_014_008.wav', 'ir_021_003.wav', 'ball_014_001.wav', 'or_021_003.wav', 'or_021_005.wav',
'or_021_010.wav', 'ball_007_002.wav', 'ball_021_003.wav', 'or_007_003.wav', 'ir_014_007.wav',
'ball_014_009.wav', 'sample_data'] ===== VIBRATION
DATA ===== Used MAT: ['109.mat', '110.mat', '111.mat',
'112.mat', '122.mat', '123.mat', '124.mat', '125.mat', '174.mat', '189.mat', '190.mat', '203.mat', '226.mat',
'229.mat', '192.mat', '136.mat', '241-2.mat'] Skipped MAT: [] Raw shape: (6720, 1024, 1) Feature shape:
(6720, 12) Classes: ['Ball_007' 'Ball_014' 'Ball_021' 'IR_007' 'IR_014' 'IR_021' 'Normal' 'OR_007'
'OR_014' 'OR_021'] First_Part_Small.csv LOADED SUCCESSFULLY Second_Part.csv LOADED
SUCCESSFULLY Third_Part.csv LOADED SUCCESSFULLY
===== CSV DATA
===== First CSV Shape : (100000, 11) Second CSV Shape:
(1888178, 26) Third CSV Shape : (94241, 11) Final fused shape: (6720, 252)
===== ACOUSTIC DATA
===== Total WAV files: 100 Audio feature shape: (100, 31)
Audio classes: ['Ball_007' 'Ball_014' 'Ball_021' 'IR_007' 'IR_014' 'IR_021' 'Normal' 'OR_007' 'OR_014'
'OR_021'] Audio Accuracy: 1.0 ===== RANDOM
FOREST ===== Accuracy: 0.6912202380952381
===== TRAINING CNN BiLSTM
===== Epoch 1/20 135/135
-----82s 565ms/step - accuracy: 0.5974 - loss: 1.2434 - val_accuracy:
0.1441 - val_loss: 2.3727 - learning_rate: 0.0010 Epoch 2/20 135/135
-----75s 556ms/step - accuracy: 0.7600 - loss: 0.6783 - val_accuracy:
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0.1329 - val_loss: 3.1468 - learning_rate: 0.0010 Epoch 3/20 135/135
 -----74s 551ms/step - accuracy: 0.7930 - loss: 0.6032 - val_accuracy:
 0.1533 - val_loss: 3.3472 - learning_rate: 0.0010 Epoch 4/20 135/135
 -----81s 545ms/step - accuracy: 0.8326 - loss: 0.4552 - val_accuracy:
 0.5428 - val_loss: 1.1944 - learning_rate: 0.0010 Epoch 5/20 135/135
 -----82s 543ms/step - accuracy: 0.8595 - loss: 0.3950 - val_accuracy:
 0.8439 - val_loss: 0.3805 - learning_rate: 0.0010 Epoch 6/20 135/135
 -----73s 542ms/step - accuracy: 0.8577 - loss: 0.3937 - val_accuracy:
 0.9052 - val_loss: 0.2832 - learning_rate: 0.0010 Epoch 7/20 135/135
 -----91s 610ms/step - accuracy: 0.8663 - loss: 0.3634 - val_accuracy:
 0.7862 - val_loss: 0.4569 - learning_rate: 0.0010 Epoch 8/20 135/135
 -----74s 545ms/step - accuracy: 0.8763 - loss: 0.3277 - val_accuracy:
 0.8058 - val_loss: 0.4400 - learning_rate: 0.0010 Epoch 9/20 135/135
 -----83s 552ms/step - accuracy: 0.8853 - loss: 0.3271 - val_accuracy:
 0.8336 - val_loss: 0.4124 - learning_rate: 0.0010 Epoch 10/20 135/135
 -----75s 556ms/step - accuracy: 0.8998 - loss: 0.2815 - val_accuracy:
 0.9154 - val_loss: 0.2102 - learning_rate: 5.0000e-04 Epoch 11/20 135/135
 -----82s 555ms/step - accuracy: 0.8907 - loss: 0.2924 - val_accuracy:
 0.6589 - val_loss: 0.9834 - learning_rate: 5.0000e-04 Epoch 12/20 135/135
 -----80s 544ms/step - accuracy: 0.9060 - loss: 0.2699 - val_accuracy:
 0.9164 - val_loss: 0.2031 - learning_rate: 5.0000e-04 Epoch 13/20 135/135
 -----83s 554ms/step - accuracy: 0.9170 - loss: 0.2366 - val_accuracy:
 0.9229 - val_loss: 0.1952 - learning_rate: 5.0000e-04 Epoch 14/20 135/135
 -----74s 546ms/step - accuracy: 0.9209 - loss: 0.2101 - val_accuracy:
 0.9145 - val_loss: 0.2078 - learning_rate: 5.0000e-04 Epoch 15/20 135/135
 -----80s 593ms/step - accuracy: 0.9291 - loss: 0.2069 - val_accuracy:
 0.9052 - val_loss: 0.2327 - learning_rate: 5.0000e-04 Epoch 16/20 135/135
 -----75s 558ms/step - accuracy: 0.9237 - loss: 0.2048 - val_accuracy:
 0.8978 - val_loss: 0.2437 - learning_rate: 5.0000e-04 Epoch 17/20 135/135
 -----75s 553ms/step - accuracy: 0.9414 - loss: 0.1666 - val_accuracy:
 0.9507 - val_loss: 0.1362 - learning_rate: 2.5000e-04 Epoch 18/20 135/135
 -----82s 550ms/step - accuracy: 0.9498 - loss: 0.1439 - val_accuracy:
 0.9433 - val_loss: 0.1545 - learning_rate: 2.5000e-04 Epoch 19/20 135/135
 -----80s 596ms/step - accuracy: 0.9467 - loss: 0.1627 - val_accuracy:
 0.9489 - val_loss: 0.1213 - learning_rate: 2.5000e-04 Epoch 20/20 135/135
 -----75s 555ms/step - accuracy: 0.9570 - loss: 0.1283 - val_accuracy:
 0.9563 - val_loss: 0.1090 - learning_rate: 2.5000e-04 42/42 -----7s
 141ms/step =====

CNN BiLSTM RESULT

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Accuracy: 0.9583333333333334

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HYBRID ENSEMBLE

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Accuracy: 0.9613095238095238 Best Model: Hybrid Ensemble

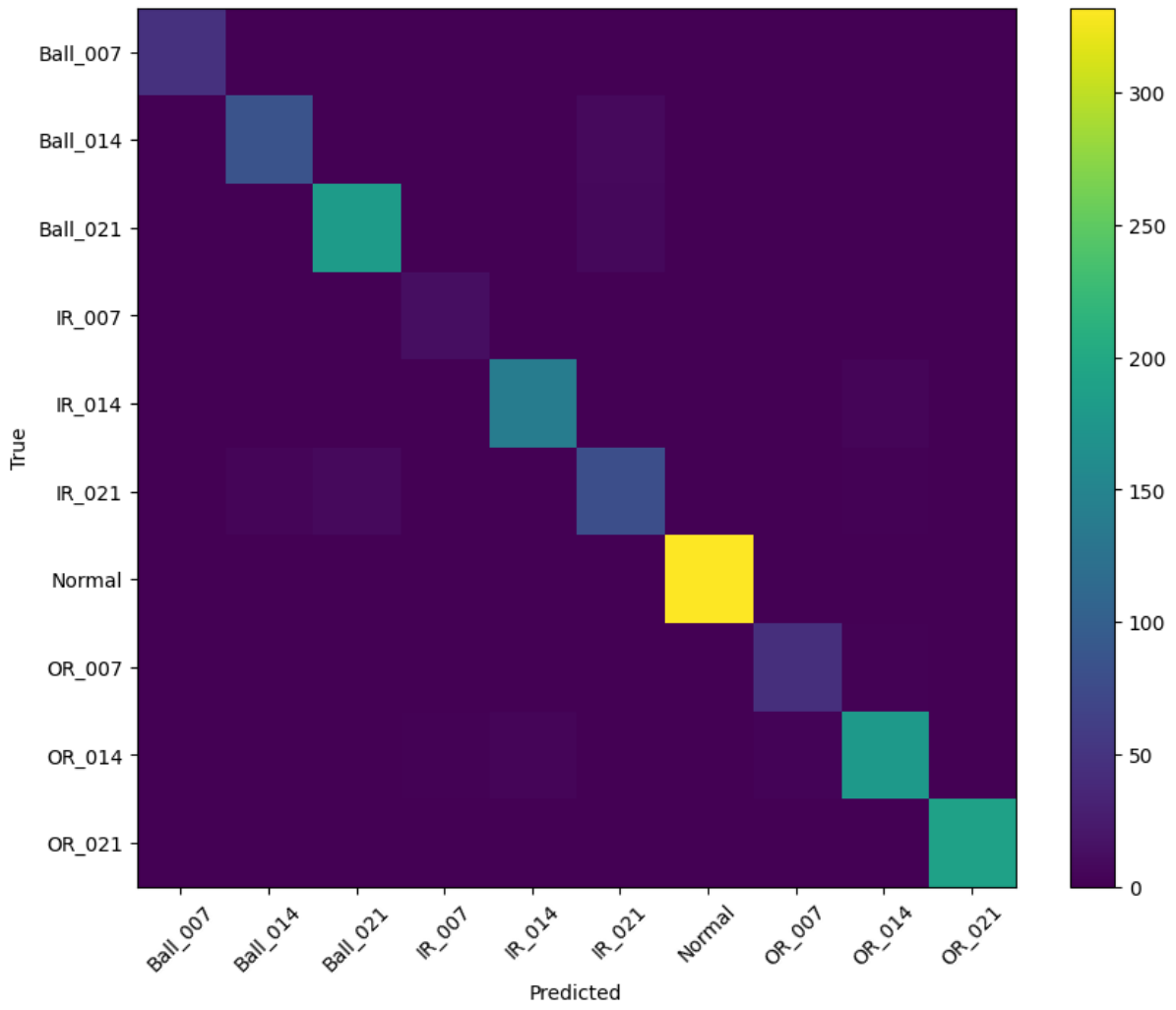
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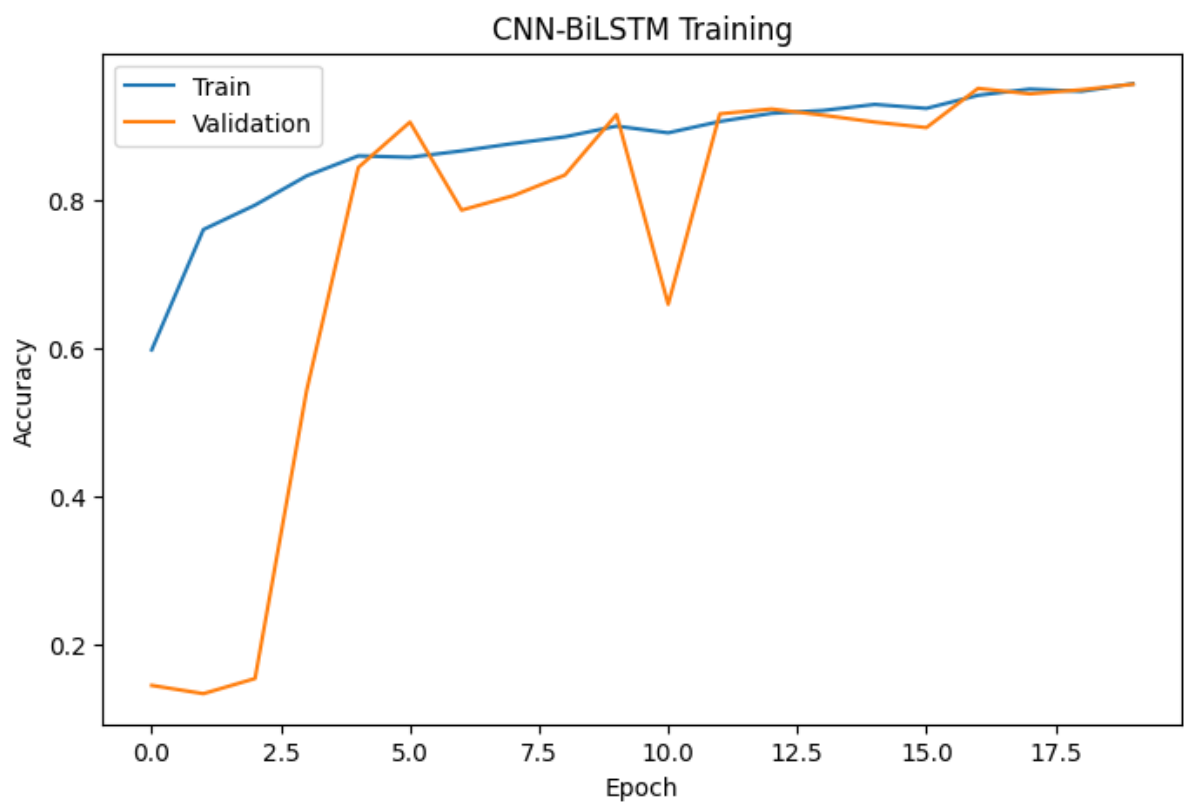
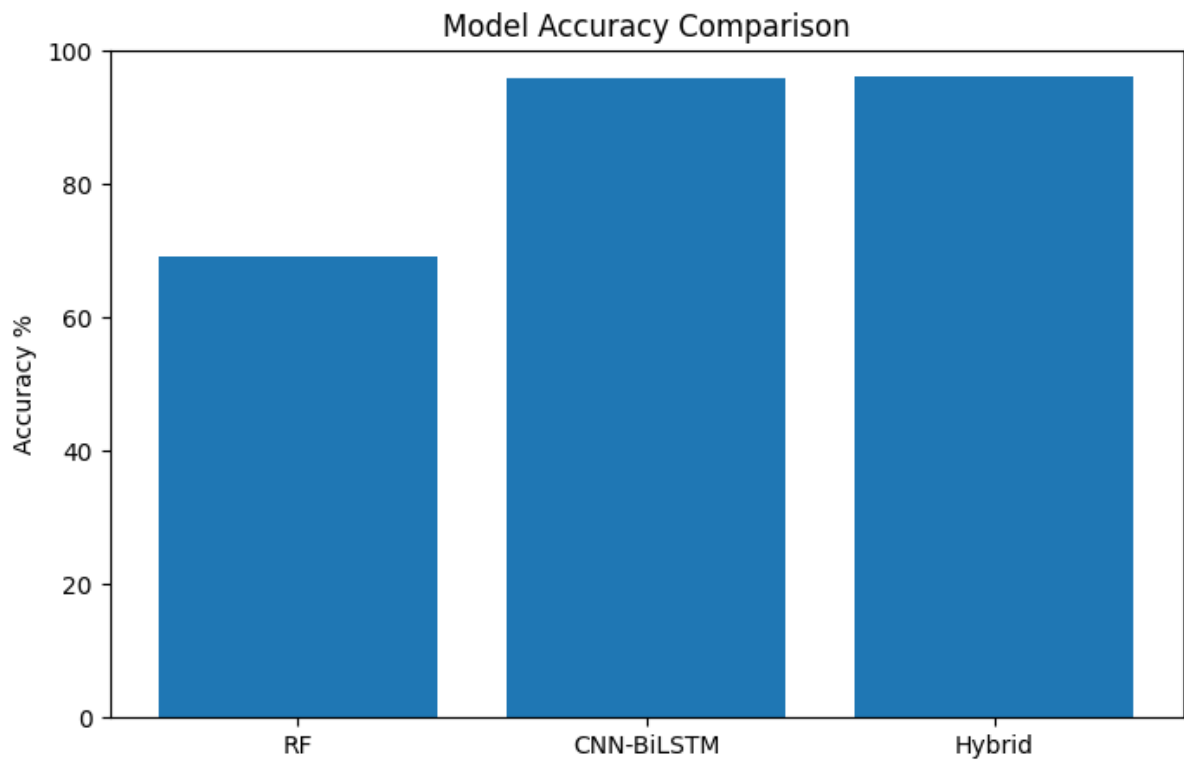
FINAL DASHBOARD

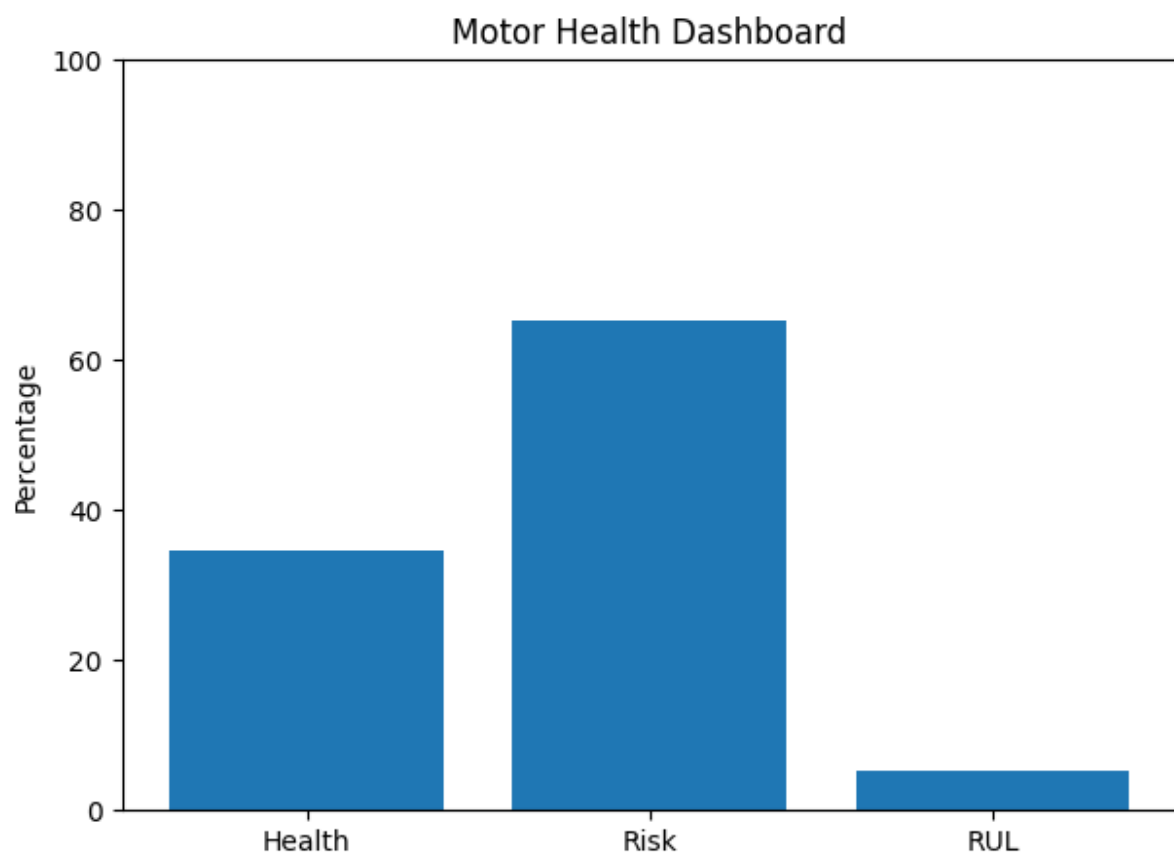
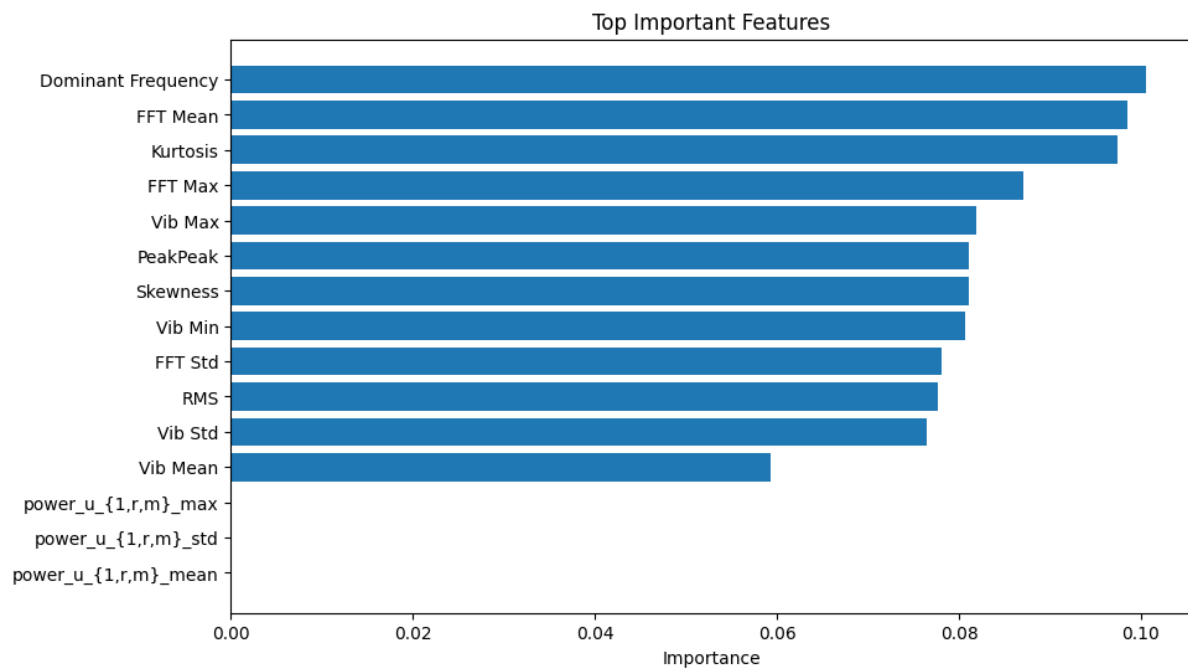
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index	Prediction Area	Output
0	Motor Type	3-Phase Induction Motor
1	Best AI Model	Hybrid Ensemble
2	Random Forest Accuracy	69.12%
3	CNN BiLSTM Accuracy	95.83%
4	Hybrid Accuracy	96.13%
5	Actual Condition	Ball_021
6	Predicted Condition	Ball_021
7	Fault Type	Ball Bearing Fault
8	Severity	Severe
9	Confidence	72.54%
10	Machine Health	34.71%
11	Risk Score	65.29%
12	Estimated RUL	5.21
13	Temperature Mean	1.17
14	Temperature Max	149
15	Current Mean	2.29
16	Current Max	35.7
17	Power Mean	2.41
18	Power Max	148.91
19	Acoustic Prediction	Ball_007
20	Acoustic Accuracy	100.0%
21	Acoustic Confidence	92.33%
22	Maintenance Status	IMMEDIATE MAINTENANCE REQUIRED
23	Recommendation	Stop motor immediately and inspect bearings.
24	Alerts	Extreme Temperature Spike, Poor Machine Health

Confusion Matrix







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SYSTEM COMPLETED SUCCESSFULLY

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ALL VARIABLES AVAILABLE YOU CAN RUN METRICS NOW

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Parameter	Value
Accuracy	0.9613
Precision	0.9616
Recall	0.9613
F1 Score	0.9614
Balanced Accuracy	0.9454
Matthews Corrccoef (MCC)	0.9548
Cohen Kappa Score	0.9548
ROC AUC Score	0.9968
Log Loss	0.2959
Sensor Fusion Score	0.8832
Predictive Maintenance IDX	0.9668
Random Forest Accuracy	0.6912
CNN-BiLSTM Accuracy	0.9583
Hybrid Ensemble Accuracy	0.9613
Acoustic Accuracy	1.0000

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CLASSIFICATION REPORT

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Class	Precision	Recall	F1-Score	Support
Ball_007	1.00	0.98	0.99	48
Ball_014	0.93	0.89	0.91	95
Ball_021	0.95	0.96	0.96	190
IR_007	0.80	0.92	0.86	13
IR_014	0.97	0.97	0.97	143
IR_021	0.83	0.84	0.84	95
Normal	1.00	1.00	1.00	332
OR_007	0.94	0.94	0.94	48
OR_014	0.95	0.95	0.95	190
OR_021	0.99	1.00	1.00	190
Accuracy	0.96	1344		
Macro Avg	0.94	0.95	0.94	1344
Weighted Avg	0.96	0.96	0.96	1344

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FINAL PARAMETERS TABLE

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No.	Parameter	Value
1	Accuracy	0.961310
2	Precision	0.961564
3	Recall	0.961310
4	F1 Score	0.961355
5	Balanced Accuracy	0.945388
6	MCC	0.954759
7	Cohen Kappa	0.954751
8	ROC AUC	0.996849
9	Log Loss	0.295858
10	Fusion Score	0.883185
11	Predictive Maintenance Index	0.966839
12	RF Accuracy	0.691220
13	CNN-BiLSTM Accuracy	0.958333
14	Hybrid Accuracy	0.961310
15	Acoustic Accuracy	1.000000

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ALL PARAMETERS GENERATED SUCCESSFULLY

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